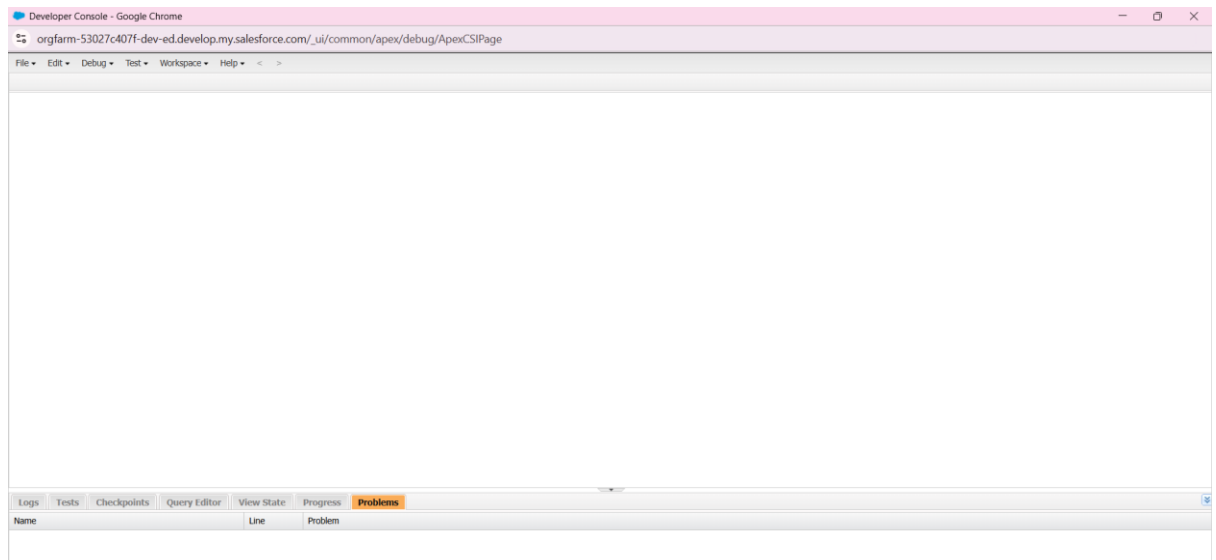


Phase 5: Apex Programming (Developer)

This phase involves custom coding using Apex Classes, Triggers, Batch Apex, Queueable Apex, Schedulers, and Test Classes. All activities are done **inside Developer Console** (no VS Code needed).

Step 1: Open Developer Console

Click on (top-right) → **Developer Console**.



Step 2: Create Apex Classes

Developer Console → **File** → **New** → **Apex Class**

Create the following classes (copy-paste code provided earlier):

1. AppointmentService – Fetch patient history & doctor schedules.
2. BillingService – Generate billing records.
3. MonthlyPatientReportBatch – Batch Apex for monthly reports.
4. AppointmentReminderQueueable – Queueable for sending reminders.
5. DailyReminderScheduler – Scheduler for daily reminder jobs.
6. AppointmentServiceTest – Test class (≥75% coverage).

📸 *Screenshots to capture:*

Developer Console - Google Chrome

orgfarm-53027c407f-dev-ed.develop.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

File Edit Debug Test Workspace Help < >

Log executeAnonymous @25/08/2025, 18:38:57 UpdateContactPhone.apxc MyfirstBatchApex.apxc AppointmentService.apxc BillingService.apxc MonthlyPatientReportBatch.apxc AppointmentReminderQueueable.apxc DailyRemi

Code Coverage: None API Version: 64 Go To

```
1 public with sharing class AppointmentService {
2
3     // Get Patient History
4     public static List<Appointment__c> getPatientHistory(Id patientId) {
5         return [
6             SELECT Id, Name, Status__c, Doctor__c, Appointment_Start_Time__c
7             FROM Appointment__c
8             WHERE Patient__c = :patientId
9             ORDER BY Appointment_Start_Time__c DESC
10        ];
11    }
12
13    // Get Doctor Schedule
14    public static List<Appointment__c> getDoctorSchedule(Id doctorId, Date apptDate) {
15        return [
16            SELECT Id, Name, Appointment_Start_Time__c, Appointment_End_Time__c, Status__c
17            FROM Appointment__c
18            WHERE Doctor__c = :doctorId
19            AND DAY_ONLY(Appointment_Start_Time__c) = :apptDate
20            ORDER BY Appointment_Start_Time__c ASC
21        ];
22    }
23 }
```

Logs Tests Checkpoints Query Editor View State Progress Problems

Name	Line	Problem
------	------	---------

Developer Console - Google Chrome

orgfarm-53027c407f-dev-ed.develop.my.salesforce.com/_ui/common/apex/debug/ApexCSIPage

File Edit Debug Test Workspace Help < >

Log executeAnonymous @25/08/2025, 18:38:57 UpdateContactPhone.apxc MyfirstBatchApex.apxc AppointmentService.apxc BillingService.apxc MonthlyPatientReportBatch.apxc AppointmentReminderQueueable.apxc DailyRemi

Code Coverage: None Go

```
1 public with sharing class BillingService {
2
3     public static Billing__c generateBilling(Appointment__c appt) {
4         Billing__c bill = new Billing__c();
5         bill.Appointment__c = appt.Id;
6         bill.Patient__c = appt.Patient__c;
7         bill.Doctor__c = appt.Doctor__c;
8         bill.Amount__c = 1000; // dummy fixed cost, change as needed
9         insert bill;
10        return bill;
11    }
12 }
```

Logs Tests Checkpoints Query Editor View State Progress Problems

Name	Line	Problem
------	------	---------

```

1 public class AppointmentReminderQueueable implements Queueable {
2
3     public void execute(QueueableContext context) {
4         // Get upcoming appointments (e.g., next day)
5         List<Appointment__c> appts = [
6             SELECT Id, Patient__c, Appointment_Start_Time__c, Appointment_End_Time__c, Status__c
7             FROM Appointment__c
8             WHERE Appointment_Start_Time__c = :Date.today().addDays(1)
9         ];
10
11         for (Appointment__c appt : appts) {
12             // Example: send a reminder (you can replace with your email logic)
13             System.debug('Reminder: Appointment for Patient ' + appt.Patient__c +
14                 ' at ' + appt.Appointment_Start_Time__c);
15         }
16     }
17 }

```

```

23 // Loop through appointments and send reminders
24 for (Appointment__c appt : todaysAppointments) {
25     if (appt.Patient__r.Email__c != null && appt.Doctor__r.Doctor_Email__c != null) {
26         Messaging.SingleEmailMessage mail = new Messaging.SingleEmailMessage();
27         mail.setToAddresses(new String[] {
28             appt.Patient__r.Email__c,
29             appt.Doctor__r.Doctor_Email__c
30         });
31         mail.setSubject('Appointment Reminder');
32         mail.setPlainTextBody(
33             'Reminder: Appointment scheduled at ' +
34             appt.Appointment_Start_Time__c.format() +
35             '\nStatus: ' + appt.Status__c
36         );
37         emails.add(mail);
38     }
39 }
40
41 // Send emails if any
42 if (!emails.isEmpty()) {

```

-
-

Step 3: Create Apex Triggers

Setup → Object Manager → Appointment__c → Triggers → New

1. PreventOverlapTrigger – Prevents overlapping appointments for the same doctor.
2. AutoBillingTrigger – Creates a Billing record when Appointment status = Completed.

Screenshot to capture:

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Details

Fields & Relationships

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Lightning Record Pages

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Compact Layouts

Field Sets

Object Limits

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Search Layouts

List View Button Layout

Triggers

2 Items, Sorted by Label

Quick Find

New

LABEL	API VERSION	SIZE WITHOUT COMMENTS	MODIFIED BY
AutoBillingTrigger	64.0	711	Shaik Jasmin, 9/24/2025, 10:11 AM
PreventOverlapTrigger	64.0	940	Shaik Jasmin, 9/24/2025, 9:07 AM

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AutoBillingTrigger

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Show Dependencies

Name	sObject Type
AutoBillingTrigger	Appointment

Code Coverage	Status
0% (0/13)	Active

Created By	Last Modified By
Shaik Jasmin	Shaik Jasmin

Created	Last Modified
9/24/2025, 10:11 AM	9/24/2025, 10:11 AM

Namespace Prefix

Apex Trigger

Version Settings

Trace Flags

```
1 trigger AutoBillingTrigger on Appointment__c (after update) {
2   List<Billing__c> billsToInsert = new List<Billing__c>();
3
4   for (Appointment__c appt : Trigger.new) {
5     Appointment__c oldAppt = Trigger.oldMap.get(appt.Id);
6
7     // Create billing record only when Appointment becomes Confirmed
8     if (appt.Status__c == 'Confirmed' && oldAppt.Status__c != 'Confirmed') {
9       Billing__c b = new Billing__c();
10      b.Appointment__c = appt.Id;
11      b.Patient__c = appt.Patient__c;
12      b.Doctor__c = appt.Doctor__c;
13      b.Amount__c = 500;
14      b.Payment_Status__c = 'Pending'; // correct field
15      billsToInsert.add(b);
16    }
17  }
```

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Apex Trigger

PreventOverlapTrigger

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Apex Trigger Detail

Edit

Delete

Download

Show Dependencies

Name	sObject Type
PreventOverlapTrigger	Appointment

Code Coverage	Status
0% (0/9)	Active

Created By	Last Modified By
Shaik Jasmin	Shaik Jasmin

Created	Last Modified
9/24/2025, 9:07 AM	9/24/2025, 9:07 AM

Namespace Prefix

Apex Trigger

Version Settings

Trace Flags

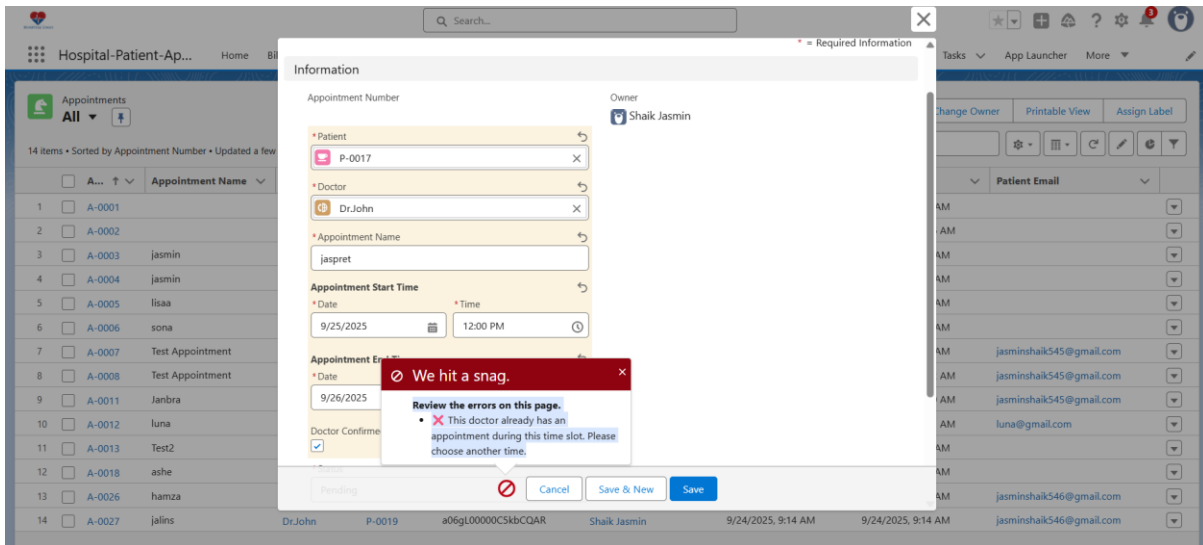
Salesforce - Developer Edition

```
1 trigger PreventOverlapTrigger on Appointment__c (before insert, before update) {
2   for (Appointment__c newAppt : Trigger.new) {
3     if (newAppt.Doctor__c != null && newAppt.Appointment_Start_Time__c != null && newAppt.Appointment_End_Time__c != null) {
4       List<Appointment__c> overlapping = [
5         SELECT Id, Appointment_Start_Time__c, Appointment_End_Time__c
6         FROM Appointment__c
7         WHERE Doctor__c = newAppt.Doctor__c
8         AND Id != newAppt.Id
9         AND (
10            (Appointment_Start_Time__c < newAppt.Appointment_End_Time__c
11              AND Appointment_End_Time__c > newAppt.Appointment_Start_Time__c)
12          )
13       ];
14       if (overlapping.isEmpty()) {
15         newAppt.addError('X This doctor already has an appointment during this time slot. Please choose another time.');
```

Step 4: Test Triggers

- Create **Appointment A**: Doctor A, 10:00–11:00.
- Create **Appointment B**: Doctor A, 10:30–11:30 → ❌ Overlap error.
- Update Appointment A → Status = Completed → ✅ Auto-generated Billing record.

📸 Screenshot to capture:

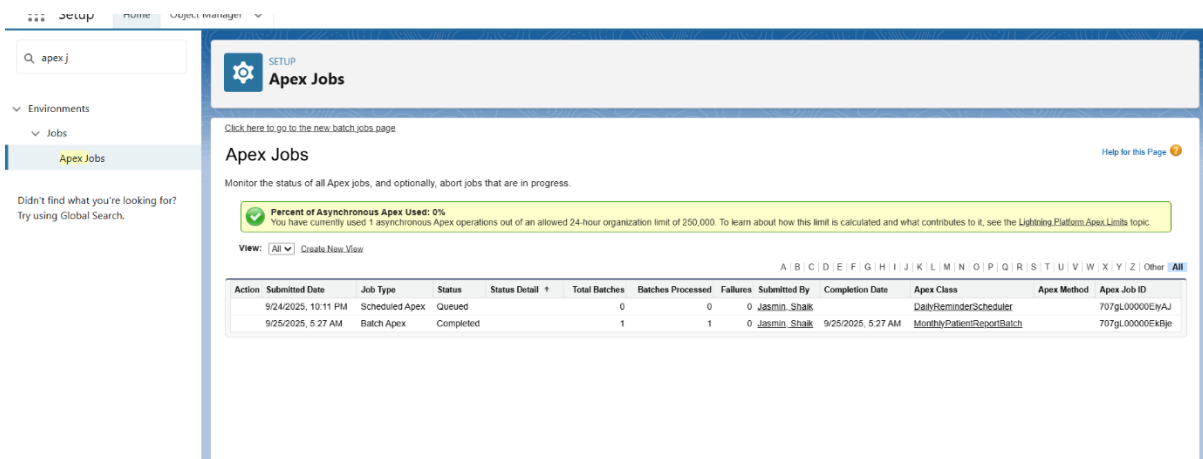


Step 5: Run Batch Apex

Developer Console → **Debug** → **Open Execute Anonymous Window** → run:

Database.executeBatch(new MonthlyPatientReportBatch(), 200);

📸 Screenshot to capture:



Step 6: Run / Schedule Daily Reminder

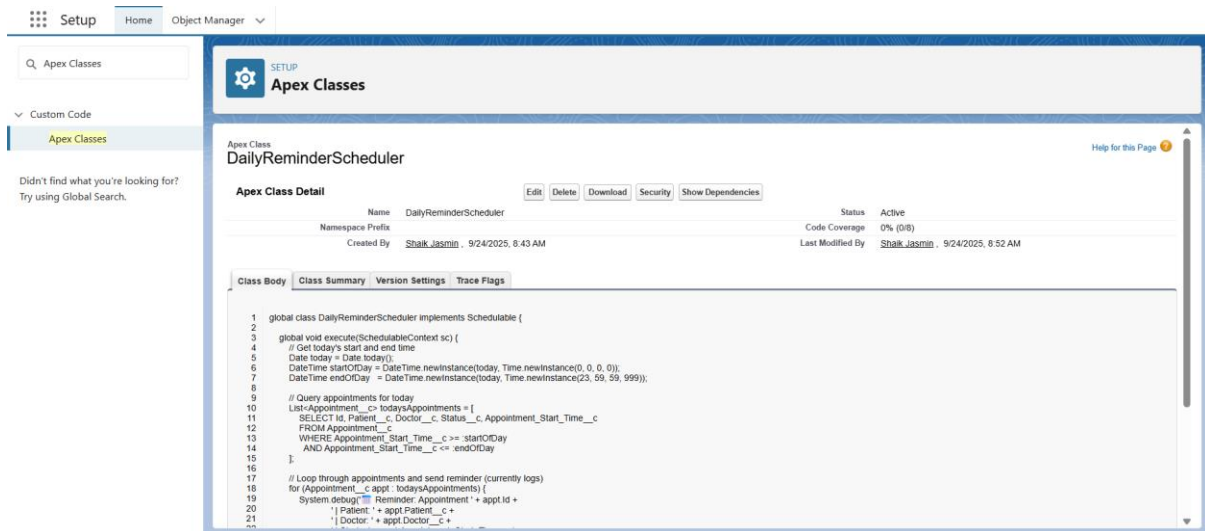
- **Run instantly:**

System.enqueueJob(new AppointmentReminderQueueable());

- **Schedule daily reminders:**

Setup → Apex Classes → **Schedule Apex** → Select DailyReminderScheduler →
Frequency = Daily @ 8 AM.

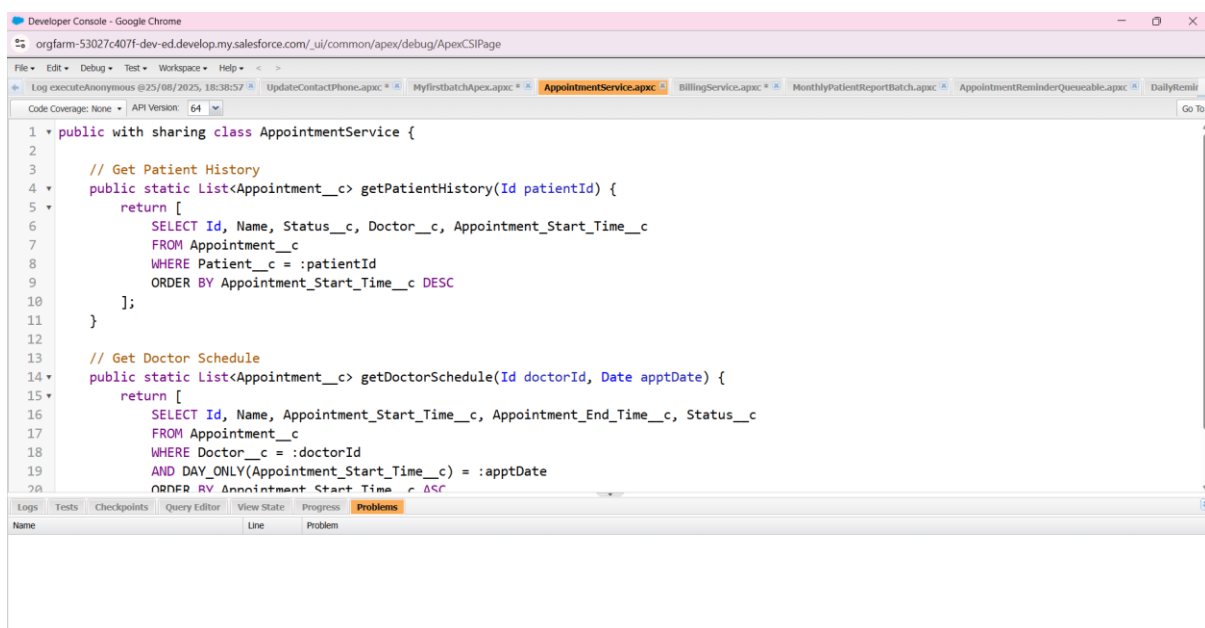
 *Screenshot to capture:*



Step 7: Run Test Classes

Developer Console → **Test** → **New Run** → Select AppointmentServiceTest.

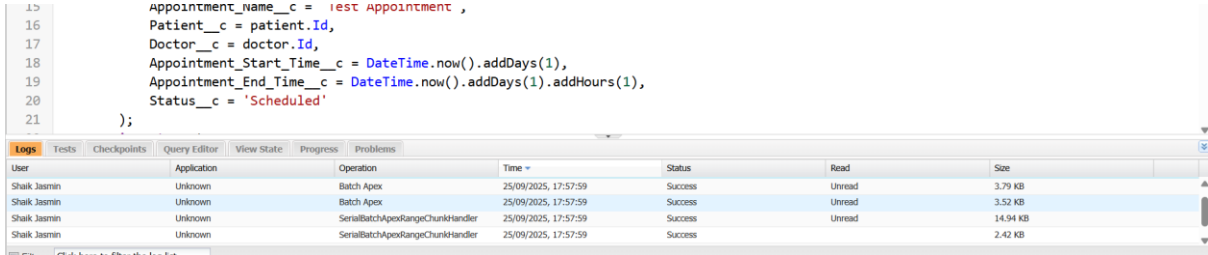
 *Screenshot to capture:*



Step 8: Verify Logs

Developer Console → **Logs Tab** → Double-click a log to open execution details.

 *Screenshot :*



The screenshot displays the Salesforce Developer Console. The top pane shows Apex code for creating an appointment. The bottom pane shows the 'Logs' tab with a table of log entries.

```
15 Appointment_Name__c = test.Appointment ,
16 Patient__c = patient.Id,
17 Doctor__c = doctor.Id,
18 Appointment_Start_Time__c = DateTime.now().addDays(1),
19 Appointment_End_Time__c = DateTime.now().addDays(1).addHours(1),
20 Status__c = 'Scheduled'
21 );
```

User	Application	Operation	Time	Status	Read	Size
Shaik Jasmin	Unknown	Batch Apex	25/09/2025, 17:57:59	Success	Unread	3.79 KB
Shaik Jasmin	Unknown	Batch Apex	25/09/2025, 17:57:59	Success	Unread	3.52 KB
Shaik Jasmin	Unknown	SerialBatchApexRangeChunkHandler	25/09/2025, 17:57:59	Success	Unread	14.94 KB
Shaik Jasmin	Unknown	SerialBatchApexRangeChunkHandler	25/09/2025, 17:57:59	Success	Unread	2.42 KB